

Automated Excel Online → PostgreSQL Data Ingestion Pipeline

Problem Statement

- Operational data was being maintained in Excel Online spreadsheets by business users.
- Analytics and reporting relied on PostgreSQL as the centralized database.
- Updates made in Excel were not automatically synchronized with PostgreSQL.
- This gap caused downstream dashboards to show outdated information unless manual database updates were performed.

Project Objective

To design and implement an automated, near real-time data ingestion pipeline that synchronizes updates from Excel Online directly into a PostgreSQL database, eliminating manual steps and improving data freshness for analytics.

System Architecture

1. Excel Online (source)
2. Power Automate (event trigger on row update)
3. FastAPI Backend API (Python REST endpoint)
4. PostgreSQL Database (central storage)
5. Power BI Dashboard (downstream consumption)

Implementation Details

Power Automate

- Configured workflow to detect row-level changes in Excel Online
- Triggered secure HTTP POST requests to the backend API

FastAPI Backend (Python)

- Developed REST API endpoint to receive and parse incoming Excel data
- Performed data validation, type checking, and schema preparation before database write

PostgreSQL Database

- Implemented UPSERT (INSERT ON CONFLICT UPDATE) logic to handle inserts and updates
- Ensured data consistency and prevented duplicate records

Deployment & Integration

- Deployed PostgreSQL instance using Docker for portability
- Enabled secure API communication for data transmission
- Monitored pipeline execution through application logs and terminal debugging

Project Results

- Achieved near real-time synchronization between Excel Online and PostgreSQL.
- Eliminated the need for manual database updates.
- Improved data consistency and reliability for downstream analytics.
- Validated the pipeline through multiple testing cycles using simulated data.

Note: This write-up is a sanitized portfolio version based on internship work. No company-specific data, credentials, or proprietary code is included.
